PROJECT REPORT

ATHLETE PERFORMANCE ANALYTICS DASHBOARD

A Centralized Java and MySQL-Based System for Athlete Performance Management

Project Focus

Centralized athlete profiles, training records, performance monitoring, workload analytics, trend analysis, personal-best tracking, squad comparison, and role-based access.

Executive Summary

The Athlete Performance Analytics Dashboard is a centralized Java-based system designed to help coaches and athletes record, monitor, and analyze sports performance. The system stores athlete profiles, training sessions, workload, heart rate, and performance metrics in a structured MySQL database. It converts raw training information into useful analytics such as training workload, Acute:Chronic Workload Ratio (ACWR), performance trends, personal bests, squad comparison, and athlete workload status.

The project addresses common limitations of paper records, spreadsheets, separate applications, and manual calculations. By bringing information into one centralized system, it reduces repetitive manual work and makes performance information easier to access, compare, and analyze.

The supplied project brief further defines the system as a tool for monitoring training and progress, analyzing workload from duration and intensity, comparing current results with previous records, identifying improvement or decline, and providing automated insights and recommendations to support coaching decisions. The overall objective is to support healthier, safer, and more effective athletic training.

Key Outcomes

- Centralized management of athlete, training, workload, and performance information.
- Role-based functions for administrators, coaches, and athletes.
- Automated workload and ACWR calculations.
- Performance trend, personal-best, and athlete/squad comparison capabilities.
- Database-backed storage using MySQL features.
- A dashboard-oriented approach that supports data-driven training decisions.

Table of Contents

1. Introduction
2. Problem Statement
3. Existing System and Limitations
4. Athlete Feedback and User Need
5. Objectives
6. Proposed Solution
7. System Workflow
8. System Architecture
9. Technology Stack
10. Features and Role-Based Menus
11. Java Program and Implementation
12. Database Design and Implementation
13. Analytics Engine
14. Results and Demonstrated Outcomes
15. Implementation Walkthrough
16. Team Contribution
17. Conclusion
18. Future Enhancement Opportunities
19. References and Source Basis

Appendix A – Project Execution Procedure
Appendix B – Key Project Statements

1. Introduction

Athlete performance management requires continuous collection and interpretation of information about training, workload, performance, and progress. When this information is maintained manually, coaches may spend significant time collecting records, calculating workload, comparing results, and looking for changes across training periods.

The Athlete Performance Analytics Dashboard addresses this management challenge through a centralized Java and MySQL DBMS-based application. The system is designed to keep athlete information and training records in one place and convert recorded data into meaningful performance analytics.

The system supports three principal user perspectives: administrators manage core records and users; coaches enter and analyze training information; and athletes view their own performance, workload, personal bests, and trends.

2. Problem Statement

Coaches often maintain athlete training information manually using notebooks or spreadsheets. This makes it difficult to track athlete performance over time, identify workload changes, compare athletes, identify performance trends, and maintain centralized records.

Core Problem

The absence of a centralized, analytics-oriented performance management system makes it harder to turn routine training records into timely information for coaching and athlete-development decisions.

3. Existing System and Limitations

The existing approach commonly relies on paper records, Excel spreadsheets, separate applications, and manual calculations. These approaches can be useful for basic record keeping, but they become less effective as the volume and variety of athlete information increases.

3.1 Existing Practices

- Paper-based recording of athlete and training information.
- Excel spreadsheets for storing and organizing performance records.
- Use of separate applications for different categories of information.
- Manual calculation of workload and other performance indicators.
- Manual comparison of present and historical athlete records.

3.2 Major Limitations

- Data duplication across files and records.
- Manual calculations that increase effort and can introduce errors.
- Difficult comparison between athletes or squads.
- Limited security and access control in basic record systems.
- No single centralized database for all performance information.
- Difficult historical analysis and trend identification.

4. Athlete Feedback and User Need

Athlete and coach feedback is an important part of validating whether a performance-management system addresses practical training needs. The project presentation identifies athlete feedback as a separate part of the project justification. However, the supplied project source does not contain individual survey responses, numerical ratings, or verbatim athlete comments.

4.1 Needs Addressed by the Project

A single place to access athlete profiles, training details, and performance records.
Easy monitoring of training and progress.
Clear workload analysis based on duration and intensity.
Ability to monitor speed, strength, endurance, and performance scores.
Comparison of current performance with previous records.
Identification of improvement, decline, or excessive workload.
Automated insights and recommendations to support coaches.
Reduced manual work and stronger support for data-driven decisions.

5. Objectives

Develop a centralized system for storing athlete profiles, training details, and performance records.
Provide coaches with efficient tools for entering and monitoring training sessions.
Calculate training workload using duration and perceived intensity information.
Calculate and display Acute:Chronic Workload Ratio (ACWR).
Monitor performance measures and identify personal bests.
Compare athletes and squads using recorded performance information.
Track historical trends and identify changes in performance or workload.
Provide role-based access for administrators, coaches, and athletes.
Use MySQL database features to maintain structured, consistent, and transaction-safe data.
Support data-driven training decisions through dashboard-based analytics.

6. Proposed Solution

The proposed solution combines Java, JDBC, MySQL Database. The Java application communicates with MySQL through JDBC, while the database provides centralized storage and supports database-level operations.
The Athletic Performance Dashboard is therefore more than a record-keeping application: it provides a structured pathway from raw training records to calculated indicators and dashboard-level insights.

6.1 Core Solution Capabilities

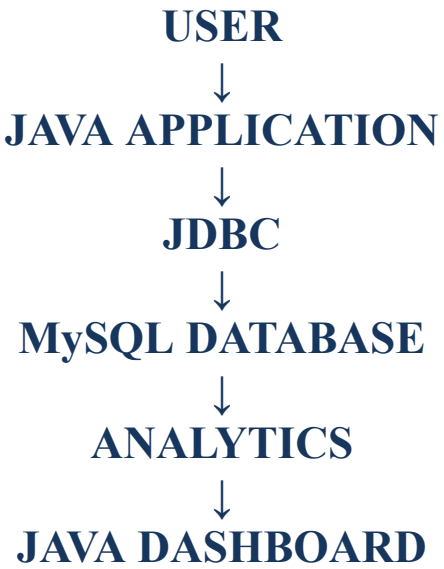
- Centralized athlete information management.
- Training-session recording and monitoring.
- Workload calculation using Duration × RPE.
- ACWR calculation using acute and chronic workload values.
- Performance trend analysis.
- Personal-best identification.
- Squad and athlete comparison.
- Athlete workload-status monitoring.
- Role-based access and menus.
- Database-backed storage and analytics.

7. System Workflow

The system follows a straightforward data-to-insight workflow. Users interact with the Java application, which communicates with the MySQL database through JDBC. MySQL operations retrieve and process the stored information, after which the analytics layer produces calculated results for the dashboard.

1	User Access	Administrator, coach, or athlete accesses the appropriate role-based menu.
2	Data Entry	Athlete profiles, squads, metrics, and training sessions are entered or maintained.
3	Database Storage	The Java application sends data to MySQL through JDBC.
4	Data Processing	operations retrieve, validate, and process the stored records.
5	Analytics	Workload, ACWR, trends, personal bests, and comparison indicators are calculated.
6	Dashboard Output	The processed information is presented to support monitoring and decision-making.

7.1 Workflow Representation



8. System Architecture

The architecture separates user interaction, application logic, database connectivity, data storage, and analytics. This structure provides a clear flow of information and supports maintainability.

Presentation / User Layer	Java Dashboard	Provides menus, forms, views, and analytics output.
Application Layer	Java	Implements application logic and user operations.
Connectivity Layer	JDBC	Connects the Java application to MySQL.
Data Layer	MySQL Database	Stores athlete, training, workload, and performance records.
Database Logic	MySQL/PL-SQL	Queries, procedures, functions, triggers, and transactions.
Analytics Layer	Workload / ACWR / Trends	Converts stored records into useful performance indicators.

9. Technology Stack

Java	Primary programming language for the application and business logic.
JDBC	Database connectivity between Java and MySQL.
MySQL Database	Centralized storage for athlete and training information.
MySQL	Data definition, retrieval, insertion, updating, and management.
MySQL/PL-SQL	Database-side procedures, functions, and related logic.
VS Code	Development environment used for project work.
GitHub	Source-code repository and project version-management platform.

10. Features and Role-Based Menus

10.1 Administrator

Manage athlete records.
Manage squads.
Manage performance metrics.
Manage users.

10.2 Coach

Add training sessions.
View athlete performance.
Calculate training workload.
View ACWR.
Compare athletes.

10.3 Athlete

View personal performance.
View personal bests.
View workload.
View performance trends.

10.4 Role-Based Access Model

Role-based menus ensure that users see functions relevant to their responsibilities. Administrators maintain core records, coaches perform training and analytical tasks, and athletes primarily consume their own performance information.

11. Java Program and Implementation

Java forms the main application layer of the project. It is responsible for user interaction, invoking database operations, and presenting the resulting information. JDBC provides the connection between the Java application and MySQL.

11.1 Typical Implementation Flow

- Initialize the Java application.
- Configure MySQL database connection details.
- Establish a JDBC connection.
- Present the appropriate role-based menu.
- Accept athlete, training, or performance information.
- Execute the required MySQL operation.
- Retrieve calculated or stored results.
- Display performance and analytics information through the dashboard.

11.2 JDBC Integration

The application uses JDBC to establish database communication. The project execution procedure specifies adding the ojdbc11.jar driver to the Java application's library, configuring database credentials, compiling the program, and running Main.java.

12. Database Design and Implementation

MySQL Database provides centralized and structured storage for the project's athlete and training data. The project demonstrates core relational-database concepts and advanced database features.

Primary Keys	Uniquely identify records.
Foreign Keys	Maintain relationships between related records.
Constraints	Enforce data integrity rules.
Normalization	Reduce unnecessary duplication and improve structure.
Views	Provide reusable query-based representations of data.
Functions	Encapsulate reusable database calculations or logic.
Stored Procedures	Perform reusable database operations.
Triggers	Automatically execute database logic in response to defined events.
Transactions	Group database operations into consistent units of work.
ACID Properties	Support reliable and consistent transaction behavior.

13. Analytics Engine

The analytics engine converts raw training information into indicators that can be used to understand workload and performance. The project specification explicitly includes training workload, ACWR, performance trends, personal bests, squad comparison, and athlete workload status.

13.1 Training Workload

Workload = Duration × RPE

In this project, workload is calculated from training duration and Rating of Perceived Exertion (RPE). This converts session duration and perceived intensity into a workload value that can be monitored across sessions.

13.2 Acute:Chronic Workload Ratio (ACWR)

ACWR = Acute Workload / Chronic Workload

Acute workload is calculated from recent training data, while chronic workload is based on a longer reference period. The ratio provides a way to compare recent training load with the athlete's longer-term load.

13.3 Performance Analytics

- Performance trends show how recorded performance changes over time.
- Personal bests identify the strongest recorded performance values.
- Squad comparison supports comparison between athletes.
- Athlete workload status helps identify workload conditions requiring attention.
- Current-versus-previous comparison helps identify improvement or decline.
- Automated insights and recommendations can be used to support coach interpretation, subject to the exact rules implemented in the final application.

Important Interpretation

The project brief defines the formulas and analytics categories, but it does not specify clinical thresholds or validated injury-prediction rules. Therefore, the system should be presented as a training-monitoring and decision-support tool rather than as a medical diagnostic system.

14. Results and Demonstrated Outcomes

The project demonstrates a centralized workflow for storing and analyzing athlete performance information. The expected demonstrated outcomes are based on the implemented features documented in the project source.

Data Management	Athlete and training information can be maintained in a centralized database.
Training Monitoring	Coaches can record sessions and view athlete performance.
Workload	Training workload can be calculated from duration and RPE.
ACWR	Acute and chronic workload values can be compared through ACWR.
Performance	Performance metrics and trends can be viewed.
Personal Bests	Athletes can view their best recorded performance values.
Comparison	Athletes can be compared and squad-level analysis can be supported.
Access	Different roles receive relevant functions through role-based menus.
Decision Support	Analytics make recorded information easier to interpret for training decisions.

No numerical improvement percentages, accuracy percentages, athlete survey scores, or injury-reduction statistics are claimed here because such measurements were not included in the supplied project material.

15. Implementation Walkthrough

A professional demonstration should follow the same logical flow as the system architecture. The following sequence can be used during the project presentation or practical evaluation.

- Start the Java application.
- Demonstrate user access and role selection.
- Show the administrator menu and basic record-management functions.
- Log in or switch to the coach workflow.

- ❖ Add or display an athlete training session.
- ❖ Show the stored training information.
- ❖ Calculate or display training workload.
- ❖ Display ACWR and explain acute versus chronic workload.
- ❖ Show athlete performance trends and personal-best information.
- ❖ Demonstrate athlete or squad comparison.
- ❖ Show the athlete view for personal performance and workload.
- ❖ Conclude by explaining how the dashboard converts stored data into decision-support information.

16. Team Contribution

The project presentation includes a team-contribution section. The supplied source does not provide individual member names or a division of responsibilities, so those details should be filled in using the team's actual contribution record.

Sajin	Programming and presentation
Morisha Maci	Report based on feedback of athletes and coaches
Sesha Sai	Analysis of workload / ACWR / performance
Mohammed Ashfaq	Documentation and Analysis

Customization Required
Replace the placeholder member rows with the actual names and responsibilities before submission. The suggested categories above are not claims about individual contributions.

17. Conclusion

The Athlete Performance Analytics Dashboard provides a centralized solution for managing and analyzing athlete performance data. It replaces fragmented manual methods such as paper records and spreadsheets with a Java-based application connected to an MySQL database.
The system stores athlete profiles, training details, workload information, heart rate, and performance records in one place. Coaches can monitor training and progress, calculate workload, view ACWR, analyze performance trends, identify

personal bests, and compare athletes or squads. Athletes can view their own performance, workload, personal bests, and trends.
The supplied project brief further establishes the dashboard as a decision-support system: it analyzes training workload based on duration and intensity, compares current performance with previous records, helps identify improvement, decline, or excessive workload, and supports automated insights and recommendations.
By integrating Java, JDBC, MySQL Database, MySql Workbench, and by demonstrating relational database features such as primary keys, foreign keys, constraints, normalization, views, functions, stored procedures, triggers, transactions, and ACID properties, the project demonstrates a complete application-and-database solution for sports performance management.

Final Outcome

Overall, the system makes athlete performance management more organized, centralized, efficient, and easier to analyze. It reduces manual work and supports more informed, data-driven training decisions, with the goal of enabling healthier, safer, and more effective athletic training.

18. Future Enhancement Opportunities

The following are logical future enhancements for a production-scale version of the project. They are recommendations rather than features confirmed as implemented in the supplied source.

- ➢ Integration with wearable devices for automatic heart-rate and activity data collection.
- ➢ Mobile access for coaches and athletes.
- ➢ Interactive charts and richer dashboard visualizations.
- ➢ Automated notifications when configured workload conditions require review.
- ➢ Export of performance reports for coaching and management review.
- ➢ More advanced predictive analytics using larger historical datasets.
- ➢ Configurable analytics thresholds based on sport, athlete, and coaching policy.
- ➢ Enhanced authentication, authorization, and audit logging.
- ➢ Cloud deployment for controlled multi-location access.

19. References and Source Basis

The following references identify the source basis and core technologies used by the project.

Project source document: Athlete Performance Analytics Dashboard, supplied with the project.

Java – programming language and application-development layer used by the project.

JDBC – Java Database Connectivity used to connect the application to MySQL Database.

MySQL Database – relational database platform used for centralized project data storage.

MySQL – database query and data-management language used by the project.

Workbench– MySQL procedural database language used for database-side programming.

VS Code – development environment identified in the project technology stack.

GitHub – repository and source-code management platform identified in the project technology stack.

For formal academic submission, institution-specific citation formatting (for example IEEE or APA) can be applied to official documentation URLs or research papers if those sources are added to the project bibliography.

Appendix A – Project Execution Procedure

The supplied project instructions specify the following basic execution sequence:

- Create the MySQL database/schema.
- Execute the SQL files in the database folder.
- Add ojdbc11.jar to java-app/lib.
- Configure database credentials.
- Compile the Java program.
- Run Main.java.

Appendix B – Key Project Statements

The following ten statements supplied for the project have been incorporated into the report's scope and conclusion:

1. The Athlete Performance Dashboard is a centralized Java-based system for managing athlete data.
2. It stores athlete profiles, training details, and performance records in one place.
3. The system helps coaches easily monitor an athlete’s training and progress.
4. It analyzes training workload based on duration and intensity.
5. Performance data such as speed, strength, endurance, and scores can be monitored.
6. The dashboard compares current performance with previous records.
7. It identifies performance improvement, decline, or excessive workload.
8. The system provides automated insights and recommendations to support coaches.
9. This reduces manual work and helps in making data-driven training decisions.
10. Overall, the system supports healthier, safer, and more effective athletic training.